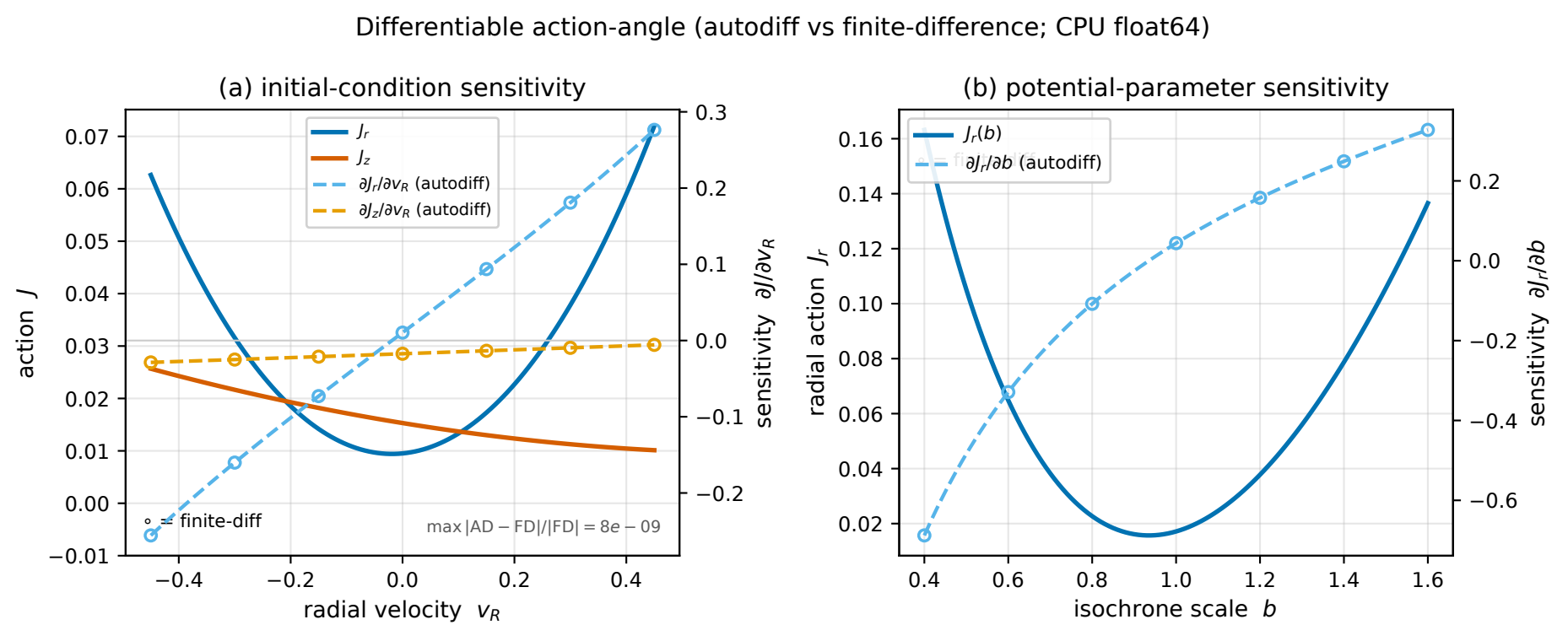
Differentiable action-angle (autodiff vs finite-difference; CPU float64)
(a) initial-condition sensitivity
J_r
J_z
$\partial J_r/\partial v_R$ (autodiff)
$\partial J_z/\partial v_R$ (autodiff)
∘ = finite-diff
max |AD − FD|/|FD| = 8e − 09
action J
sensitivity $\partial J/\partial v_R$
radial velocity v_R
(b) potential-parameter sensitivity
$J_r(b)$
$\partial J_r/\partial b$ (autodiff)
radial action J_r
sensitivity $\partial J_r/\partial b$
isochrone scale b